Estimation

Exam code: 0607

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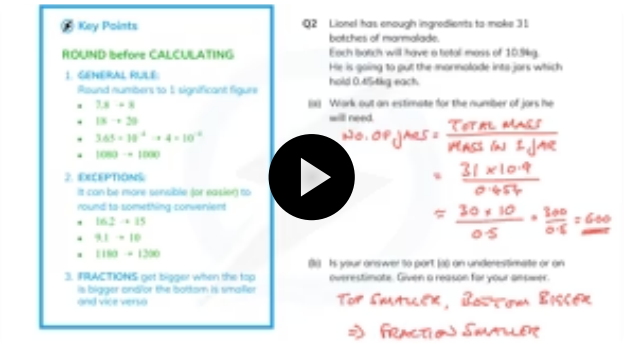

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Estimation

Why do I need to estimate?

- **Estimation** can be used to find **approximations** for difficult calculations
- You can estimate a calculation to **check your answers**
  - You can identify if there is a mistake in your working out if your answer is much bigger or smaller than your estimated value

How do I estimate?

- **Round** each number in the question to something sensible then perform the **calculation**
  - The exam question will usually tell you what to round each number to before carrying out any calculations
- The **general rule** is to round numbers to **1 significant figure**
  - $7.8 \rightarrow 8$
  - $18 \rightarrow 20$
  - $3.65 \times 10^{-4} \rightarrow 4 \times 10^{-4}$
  - $1080 \rightarrow 1000$
- In certain cases it may be more **sensible** (or easier) to round to something convenient
  - $16.2 \rightarrow 15$
  - $9.1 \rightarrow 10$
  - $1180 \rightarrow 1200$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

underestimated or overestimated?

- For **addition** $a + b$ and **multiplication** $a \times b$

a (rounded **up**) and/or ***b*** (rounded **up**)	Overestimate
a (rounded **down**) and/or ***b*** (rounded **down**)	Underestimate

- For **subtraction** $a - b$ and **division** $a \div b$

a (rounded **up**) and/or ***b*** (rounded **down**)	Overestimate
a (rounded **down**) and/or ***b*** (rounded **up**)	Underestimate
a (rounded **up**) and ***b*** (rounded **up**)	Not easy to tell
a (rounded **down**) and ***b*** (rounded **down**)	Not easy to tell

Examiner Tips and Tricks

- Estimation exam questions often involve small decimals
  - Avoid rounding to 0, especially if the small decimal is the denominator of a fraction, as dividing by 0 is undefined

Worked Example

Calculate an estimate for $\frac{17.3 \times 3.81}{11.5}$. State, with a reason, whether the estimate is an overestimate or an underestimate.

Answer:

Round each number to 1 significant figure

$$17.3 \rightarrow 20$$
$$3.81 \rightarrow 4$$
$$11.5 \rightarrow 10$$

Perform the calculation with the rounded numbers

$$\frac{20 \times 4}{10} = \frac{80}{10} = 8$$

An estimate is 8

This is an overestimate as the numerator was rounded up and the denominator was rounded down

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